

Review

A Review of Organic Waste Treatment Using Black Soldier Fly (*Hermetia illucens*)

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Abstract: The increase in solid waste generation is caused primarily by the global population growth that resulted in urban sprawl, economic development, and consumerism. Poor waste management has adverse impacts on the environment and human health. The recent years have seen increasing interest in using black soldier fly (BSF), *Hermetia illucens*, as an organic waste converter. Black soldier fly larvae (BSFL) feed voraciously on various types of organic waste, including food wastes, agro-industrial by-products, and chicken and dairy manure, and reduce the initial weight of the organic waste by about 50% in a shorter period than conventional composting. The main components of the BSFL system are the larvero, where the larvae feed and grow, and the fly house, where the adults BSF live and reproduce. It is essential to have a rearing facility that maintains the healthy adult and larval BSF to provide a sufficient and continuous supply of offspring for organic waste treatment. The BSF organic waste processing facility consists of waste pre-processing, BSFL biowaste treatment, the separation of BSFL from the process residue, and larvae and residue refinement into marketable products. BSFL digest the nutrients in the wastes and convert them into beneficial proteins and fats used to produce animal feed, and BSFL residue can be used as an organic fertilizer. This review summarizes the BSFL treatment process to provide an in-depth understanding of the value of its by-products as animal feed and organic fertilizer.

Keywords: bioconversion; black soldier fly; composting; organic waste management



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1. Introduction

Population growth brings a higher standard of living that results in direct or indirect solid waste generation. Solid wastes are generated by the agricultural, commercial, construction, hazardous, industrial, and food sectors [1]. The generation of solid waste is projected to increase with the rapid growth of the global population. A United Nations report projected that the current world population of about 7.6 billion will reach 8.6 billion in 2030 and 9.8 billion in 2050. The recent decades have seen significant changes in mass production, mass consumption, and mass waste disposal due to metropolitan area network spread, economic development, and consumerist ways of life [2,3]. These factors contribute to the continuous generation of large volumes of wastes. According to Kaza et al. [4], about 1.3 million tons of solid waste are generated daily. The amount is projected to increase from two billion tons in 2016 to 3.4 billion tons in 2050. There is an urgent need for researchers to develop better technologies and processes to deal with the problems associated with waste generation.

The primary objective of waste management is to reduce the amount of generated waste and, thus, reduce the disposal cost and the impacts on the environment and on human health [5]. A poor waste management system will lead to the environmental issues typically experienced by most low- and middle-income countries (LMIC) due to the tactless open dumping and uncontrolled burning that cause air and water pollution [4]. Between 50–70% of the solid wastes dumped in landfills are organic materials, and the remaining are non-organic materials comprising a mixture of plastics, metals, paper, and glass. Waste pickers collect large amounts of non-organic waste for recycling and leave a large proportion of organic wastes to decompose at the dumpsites [6]. The decomposition of organic waste emits methane and carbon dioxide as end-products and causes the accumulation of greenhouse gases that contribute to climate change [7]. In addition, the unhygienic conditions at landfills encourage the breeding of vermin, flies, and rodent vectors that spread diseases such as cholera and malaria in the community [8]. Composting is one of the methods that can reduce the amount of organic waste sent to landfills.

Composting is a proven technology for treating organic waste and can significantly reduce the amount of generated waste. The conversion of organic refuse by saprophage (CORS) systems, which feeds decaying organic waste to organisms (saprophages), can improve composting performance. The well-known application of CORS is vermicomposting, in which worms and microorganisms convert organic waste into nutrient-rich humus [9]. The black soldier fly (BSF), *Hermetia illucens* Linnaeus (Diptera: Stratiomyidae), have been introduced as organic waste converters. Researchers have focused on a relatively new organic waste treatment method using BSF [10,11]. BSF larvae (BSFL) voraciously consume organic-rich waste such as food wastes [12], agro-industrial by-products [13], and dairy manure [14]. Thus, the nutrients in BSFL can be converted into essential proteins and fats used in animal feed [15,16], and fulfill the shortage of conventional animal feed, the price of which has been increasing over the years [17]. Moreover, the residue from the BSFL bioconversion process can be used as a fertilizer [16,18]. Another benefit of this process is that BSF adults are non-pathogenic, since they do not feed. Researchers found that BSFL secrete harmful bactericidal compounds that prevent house flies from laying eggs and diminish foodborne pathogens, such as *Escherichia coli* and *Salmonella enterica*. Hence, there is no concern with the transmission of diseases due to BSF farming on an industrial scale.

The economic feasibility of waste management is dependent on several factors, including the type of organic source, nutrient content, and the conversion ratio of waste to biomass [19]. BSFL's conversion of organic waste is an excellent recycling technique and a comprehensive solution because of its fewer adverse environmental impacts, such as less greenhouse gas emissions and smaller ecological footprint for producing protein feed and other nutrient supplements. The low capital investment of this method makes its implementation feasible for low- and middle-income countries [20]. Much research has been conducted on BSF larvae composting. This paper will summarize the findings of selected previous studies on BSFL organic waste treatment and the potential use of BSF by-products that focused on larvae and frass, thereby providing a comprehensive and concise source of knowledge to the reader.

2. The Life Cycle of Black Soldier Fly

The black soldier fly (BSF), *Hermetia illucens*, is a true fly (Diptera) of the family *Stratiomyidae* that has a promising potential as a cost-effective alternative for recycling biological waste [21]. Even though the BSF originates in the tropical, subtropical, and northern regions of the Americas, they are now present in tropical and temperate areas of the world [22,23]. BSF adults are not strong fliers and spend most of the day resting on plants, and live approximately two weeks only on water. Unlike other flies, BSF adults do not bite and ingest food because they do not have a stinger, mouthpart, or digestive organs. They are 15–20 mm long, and the male BSF has a bronze abdomen, while the female is reddish-brown [24]. The adult female BSF mates and oviposits only once in her lifetime. The BSF

is a eurygamous insect that mates during flight and requires broad areas for their nuptial flight [25].

The male BSF emerges two days earlier than the females, and mating occurs after two days. The females lay eggs in crevices in dry areas near the larval substrate; the eggs hatch in about four days into a neonate larvae stage [26]. The larvae grow on organic waste, including manure, food scraps, municipal refuse, and decomposed plant material [22,27]. In the final stage, the larvae can reach 27 mm in length and widths of up to 6 mm, and weigh up to 220 mg [28]. The prepupae, the last larval stage before pupation, empty their digestive tract and leave the food source to search for a dry and safe spot to pupate; this behavior is the “self-harvesting” that eliminates the previously labor-intensive stage of their farming [22]. The larvae have now reached their maximum size and have a 36–48% protein and 33% fat content [29], and will develop into adults in about 14 days [30,31]. Figure 1 shows the life cycle of the BSF, from egg to adult, which takes an average of 40–43 days in the tropics [26]. However, the cycle could take up to four months [32] since the larval and prepupal stages may extend, depending on food availability and other conditions. BSFL feed on as much organic waste as they can and stop feeding once they pupate and transform into adults. The fat reserved in their body is converted into energy for adults’ mating and continuous life cycle [26]. Nevertheless, environmental conditions, such as temperature, light, and humidity, are the critical factors for ensuring a successful BSF life cycle, which will be discussed further in the sub-topic on the BSF treatment system.

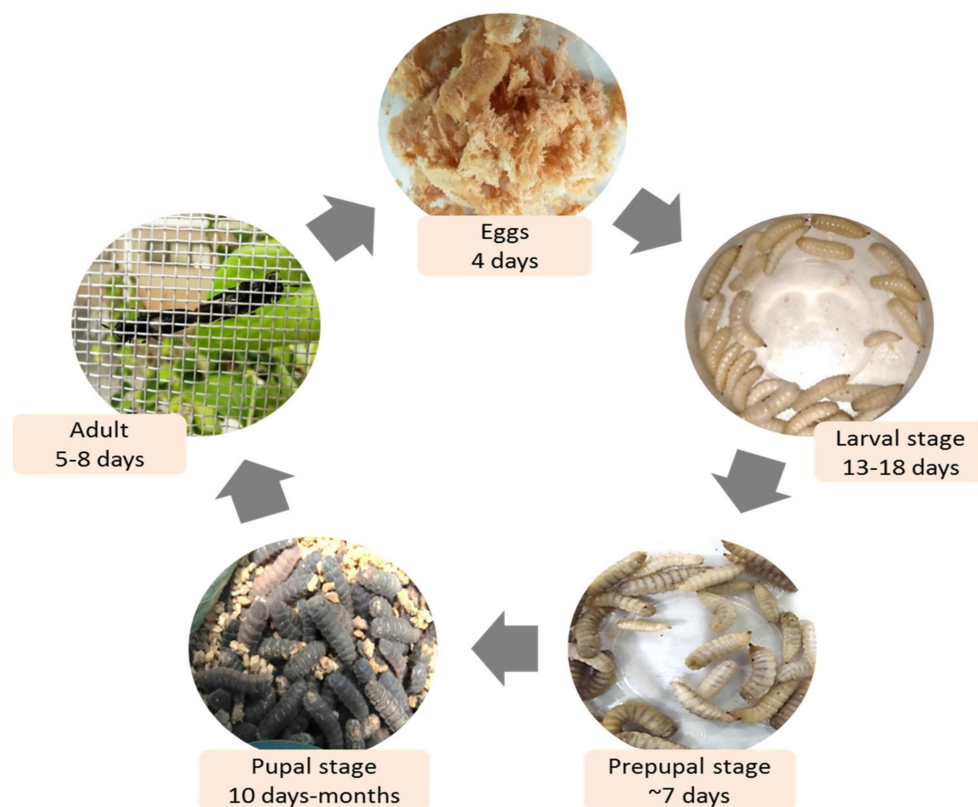


Figure 1. The life cycle of the black soldier fly, *Hermetia illucens*.

3. Organic Waste Treatment via Black Soldier Fly Farming

The unique characteristic of BSFL is that they can consume large amounts of organic wastes for growth until they reach the prepupae stage [33]. Compared to conventional composting, BSFL are more effective in reducing 50% of the organic waste in a shorter period. Because of this, researchers have focused on using BSF as a sustainable alternative for treating organic waste. Table 1 lists the different types of organic waste used in BSF waste treatment. Organic waste treatment with BSF is cost-effective and emits less pollution [34].

According to Diener et al. [35], large-scale facilities use BSFL to produce protein and treat up to 200 tons of waste each day. However, the details of the current design and operating procedures for effective large-scale BSF treatment is commercially sensitive information and not shared [36]. This section presents a brief review of the systems and designs of the BSF waste treatment used by previous researchers.

Table 1. BSF waste treatment studies using different types of organic waste.

Substrate	Subject	Reference
Food waste	Carbon and nitrogen conversion in food wastes by BSFL. Effects of moisture/water content on larval growth and composting efficiency.	[37] [38,39]
Fruit and vegetable waste	The potential of fruit and vegetable waste as rearing media for BSFL.	[40]
Poultry feed	The BSFL nutrition composition changes throughout its life cycle.	[15]
Human manure	The efficiency of BSFL bioconversion in human fecal waste treatment. The reduction of pathogenic microorganisms in human feces by BSFL.	[41] [42]
Animal manure	Use of BSFL in chicken manure management.	[43,44]
	Effects of companion bacteria when BSFL converts chicken manure (CHM) into insect biomass.	[15,43]
	Comparison of the suitability of different manures as a feeding substrate for BSFL.	[45,46]
	Bioconversion of dairy manure by BSFL.	[14,47,48]

3.1. Black Soldier Fly Organic Waste Processing System

The BSF organic waste processing facility consists of waste pre-processing (e.g., particle size reduction, dewatering, and inorganic waste removal), BSFL biowaste treatment, separation of BSFL from the process residue, and larvae and residue refinement into marketable products [49]. Figure 2 shows the flow of a basic BSF treatment process. The system comprises the larvero, which is a place where the larvae grow, while the fly house is where the adults live and reproduce [21]. A rearing facility that maintains healthy adult and larval BSF is essential to ensure a sufficient and continuous supply of offspring for organic waste treatment [49]. Larvae and frass are the by-products produced in biomass by the end of the BSF treatment process, which can be converted into animal feedstuff and organic fertilizer. Overall, BSFs are particularly compelling, since they offer a natural and cost-effective potential alternative for recycling biological waste.

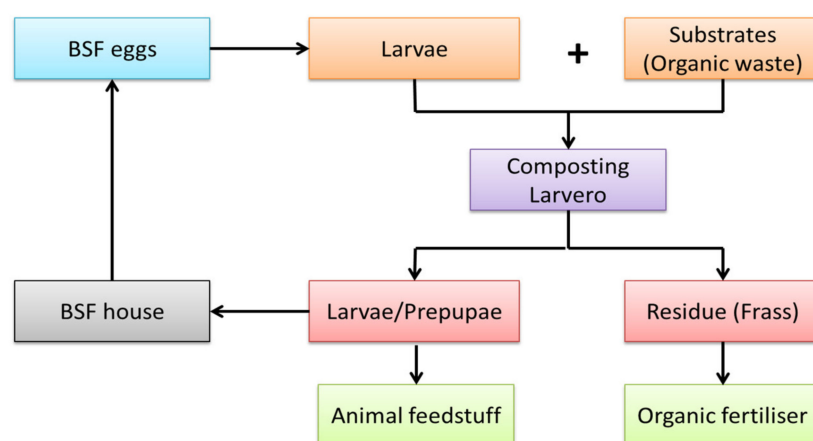


Figure 2. A flow of a basic BSF treatment process.

3.1.1. Black Soldier Fly Egg-Trapping

The initial step of BSF treatment is to purchase the BSF eggs from the market or capture them in the wild. The major components in the egg-trapping are baiting and trapping materials. The egg-trap set-up to acquire BSF eggs in the wild consists of a rearing box, the substrate as a food source (any organic material, such as fruits, vegetables, or animal

manure), a nylon net, and corrugated cardboard pieces. Booth and Sheppard [50] used chicken manure as bait and placed the corrugated cardboard on chicken manure as a trap for the female BSF to lay the eggs in the flute of the corrugated cardboard. The nylon net allows the odor to escape while preventing parasitic and predatory insects from entering and disrupting the system [51]. The rearing box containing the substrates is placed in the open to attract female BSFs.

Previous studies showed that female BSFs laid eggs on various organic matters. Sripon-tan et al. [51] investigated the use of several substrates to attract females and found more egg clutches near the fruit waste traps than animal manure. Ewusie et al. [52] found that piggery dump waste is a more suitable site to trap BSF eggs than poultry and sheep waste microhabitats. This study found that, even though female BSFs are attracted to various types of decaying organic waste, they tend to choose those with higher nutrition content for their offspring. In addition to the baiting materials, environmental factors influence the efficiency of BSF egg-trapping. The vegetation growing around the site, for example, is an ideal trapping site because it provides a resting area for the adults and protects them from heat and rain [53]. Placing the rearing box at a desirable site will increase the egg-trapping efficiency. The corrugated cardboard where the females deposited their eggs is placed in the larvero system [54]. The hatched larvae are reared until they are 4–6 days old (DOL), and are then placed in the BSFL treatment unit. The organic waste for BSFL feeding substrates is pre-treated to optimize BSFL conversion performance [14].

3.1.2. Pre-Treatment of the Black Soldier Fly System

The pre-treatment of the substrate is critical to improving substrate biodegradability/digestibility, making it easier for the BFSL to digest the substrate. It also promotes the growth of the BFSL. The capacity of BSFL to degrade waste in a specific time can be evaluated by using the waste reduction index (WRI, Equation (1)). Higher WRI values indicate a good reduction efficiency [9].

$$\text{WRI} = [(W - R)/W]/t \times 100 \quad (1)$$

where W is the total quantity of feeding substrate used during the time t , and R is the residue left at harvesting time t .

Dortmans et al. [49] recommended shredding or grinding the substrate after separating the inorganic materials from the hazardous materials because the larvae take longer to break down large chunks of the substrate due to their small mouthparts. The shredding or grinding also increases the surface area of the substrate, which promotes the growth of beneficial microbes and homogenizes the substrates, which improves the consistency of nutrient availability in the mixtures [55].

Several studies have investigated the pre-treatment methods for improving substrate biodegradability and nutrient viability [14]. Gao et al. [56] used shredded maize straw as a substrate; the optimal fermented straw was obtained by mixing *Aspergillus oryzae* in an inoculation ratio of 4000:1 and fermenting it for 24 h in a 32.3% water content. Wong and Lim [57] investigated the self-fermentation of waste coconut endosperm and discovered that the optimum duration to increase the nutrition content in waste coconut endosperm is four weeks. These studies added microorganisms to the substrate to increase the hydrolysis rate of lignocellulosic materials [33]. Researchers also recommended pre-treating by mixing the waste with other substrates to obtain a balanced nutrient content that ensures larval growth. For example, researchers have added rice straw with glucose to restaurant waste [58] and dairy manure to soybean curd residue [14]. Xiao et al. [16] found that mixing sewage sludge with chicken manure shortened the larval development time from 39 days to 12–13 days, compared to using only a sewage sludge mixture. This study shows that blending two or more substrates could enhance the nutritional value, which ensures larval growth and, thus, helps the BSF complete its life cycle in a shorter time. In summary, pre-treatment is one of the factors determining the effectiveness of a BSF treatment system.

3.1.3. Black Soldier Fly Treatment System

A particular concern with BSFL organic waste treatment is its varying reliability and efficiency [59]. Researchers used various types of organic waste as a substrate in BSF treatment, including kitchen waste [12], poultry waste [60], dairy manure [48], and human feces [42]. Tinder et al. [61] contended that the macronutrients in organic wastes, such as protein, carbohydrates, fibers, and lipids, have a considerable influence on the process performance. Protein is an essential nutrient in larval feeding substrates because of its significant positive effect on larval development [62]. Nguyen et al. [55] and Oonincx et al. [45] found that BSFL reared in organic wastes with higher protein contents have higher larval weights, larval proteins, bioconversion rates, feed conversion rates, and shorter development times and lower lipid contents. Jalil et al. [63] have proven that larvae reared in protein food sources are larger than those feeding on a carbohydrate food source. Jucker et al. [40] found that larvae feeding on substrates with low protein contents have a longer development time, are smaller, and have higher lipid contents if the carbohydrate content is high [40]. The different nutrient contents of the organic wastes determine the BSFL's performance. For example, municipal organic solid wastes have the highest lipid contents; municipal organic solid waste and animal manure have higher protein contents than fruits and vegetables, which have higher carbohydrate contents. Animal manures and fruit and vegetable wastes have a higher median fiber and ash content [59]. Therefore, enhancing and balancing the nutritional value of the BSFL feeding substrates by blending several nutrient-rich substrates during the pre-treatment step can improve the process performance and quality of BSFL biomass. The efficiency of BSFL treatment is dependent on the type, quantity, and quality of feed, and also various environmental factors.

Previous studies have determined the optimum range for the operating parameters for process performance. BSFL can live in aggressive environmental conditions, such as drought, oxygen scarcity, and food shortages, to a certain degree. Nonetheless, the ideal temperature range for BSF adults' mating and oviposition is between 27–37 °C, 40–60% humidity, and 60–70% relative humidity [64]. Lalander et al. [38] studied the BSF composting process efficiency by providing ventilation and found that 80–90% water content is adequate for BSFL feeding substrate. Cheng et al. [39] investigated the trade-off between larval growth rate and residue sieving efficiency. They found that even though a higher moisture content of the food waste results in a higher larval growth rate, it is difficult to separate the residue from the BSFL biomass. They recommended an optimum range of 70–75% for efficient residue sieving because sieving is no longer feasible at a moisture content of more than 80%. The studies on substrate pH have determined the impact of pH on larval development. Ma et al. [18] found that both acidic (pH = 2, pH = 4, and pH = 6) and basic (pH = 8 and pH = 10) conditions contribute to high BSFL survival. However, a pH range of 6 to 10 is more conducive for optimal larval growth and higher larval weight relative to a pH range of 2 to 4 because the alkalization of the substrates by larval activity stimulates protease activity, which increases the amount of protein available for larval growth [65]. The BSFL bioconversion process increases the pH of the substrate and leaves an alkaline residue, making the substrate suitable for use as fertilizer; the ideal pH range of 7 to 8 stimulates plant growth and provides a favorable environment for nurturing beneficial bacterial populations in the residue [25,66].

The substrates and neonate larvae (4–6 DOL) were placed in the larvero of the BSFL treatment unit to begin the composting process. The young larvae feed on the substrate and grow while processing and reducing the waste. The number of BSFL added to the substrate is dependent on the amount of waste in a specified volume and surface area [49]. Lalander et al. [19] have shown that the efficiency of the operation is influenced by larval density, feeding rate, and feeding mechanism. The ideal condition for an experimental setting is a larval density of 1.2 larvae/cm² and a feeding rate of 163 mg/larva/day (dry base), which yielded up to 1.1 kg/m²/day of larval compost and 59 g/m²/day of larval biomass on a dry basis [67]. The BSFL can be fed by a batch (TFS) or daily (DFS) feeding system. Meneguz et al. [13] found that TFS larvae have a shorter development time, but DFS larvae

have a higher final weight. The thickness of the substrate layer in the larvero is limited to no more than 5 cm. The larvae cannot process the entire layer if the substrate is more than 5 cm thick, and the unprocessed substrate will rot and produce an odor that attracts other filth flies. The composting process in the larvero continues until the larvae grow large enough for harvest after 14–18 days of feeding. The prepupae crawled out on their own and had the advantage of being already separated from the residue. A high portion of the prepupae remained in the substrate, which resulted in an unwanted fly population and a loss of harvest [49]. Therefore, it is essential to sieve the substrate to separate the BSFL biomass from residue before refining the marketable products.

3.1.4. Post-Treatment of the Black Soldier Fly System

The main by-products of the BSFL organic waste treatment are the larvae and frass that need to be separated to obtain valuable products, such as animal feedstock and fertilizer. BSFL accumulates adequate nutrients, such as proteins and lipids, during the larval stage, since this is the only feeding duration in the BSF lifecycle. The larvae then undergo pupation and emerge as adult flies, leaving a residue called frass. The main advantage of the BSF system is that the larvae can consume a variety of organic materials for growth, including decomposable by-products and wastes, until they reach the prepupae stage [33]. Diener et al. [27] recommended adding a food supply to the larvero daily, providing a drainage system to drain the excess leachate, and a ramp for the prepupae to crawl out from the larvero, which also ensures the easy separation of the pupae from the frass [27].

The sieving is conducted in the final stage to refine the residue quality since many prepupae are still present in the material [49]. In the conventional approach of BSFL bioconversion, the organic waste is fed directly to the larvae without any moisture adjustment [27]. This method is straightforward and saves time; however, it is difficult to separate the residue from BSFL biomass when the residue is excessively wet with a moisture content of 82–86%, making it too viscous for sieving [30]. Proper moisture control can overcome this problem. Cheng et al. [39] conducted an experiment to determine the effect of food waste moisture contents of 70%, 75%, and 80% on residue separation. The ideal moisture contents for sieving food waste are 70 and 75%. Residue with 80% initial moisture content is not suitable for sieving because the adhesive property of the water molecules promotes particle aggregation, which increases the particle size. Dortmans et al. [49] suggested harvesting, which is the process of separating the larvae from the residue using a manual or automated shaking sieve after 12 days of BSFL waste treatment, when the larvae have reached their maximum weight and nutritional value but have not developed into prepupae. The larvae or prepupae can be processed to produce animal feed, and the residue is refined to produce organic fertilizer.

4. BSF–Organic Waste Processing Products

The extracted protein meal and oils, and other by-products from BSFL treatment, can be used as animal feed, biodiesel, and fertilizer [68]. BSF is an insect with a short lifecycle, rapid growth, and promising potential. BSF farming is a new approach for recycling organic waste, such as animal waste [55], kitchen waste [31], and other types of organic waste, and converting the low-value wastes into high-value, marketable ingredients [64]. BSFL are safe for consumption by animals because they are pest-free and do not accumulate mycotoxins or pesticides in their bodies. However, this production approach has a social stigma that forbids the consumption of organisms [34]. Thus, researchers have conducted studies to determine the compositions and effects of BSFL biomass and residue, in general, to educate the public about the BSFL processing products.

Figure 3 shows the general flow of processing raw wastes to obtain the by-products in the BSF waste treatment system. Based on Figure 3, the utilization of *H. illucens* flies is promising where the larvae are a source of high-quality protein and, on the other hand, can consume organic waste throughout their lives, consequently reducing the disposal waste. However, when these flies are bred on a large scale, secondary waste is produced,

leaving large numbers of dead fly biomass. Ushacova et al. [69] found that this biomass has the potential to be an important renewable source of pigments such as melanin and ommochrome. Overall, all biomass produced throughout BSF organic waste system is useful, and can be converted into valuable products.

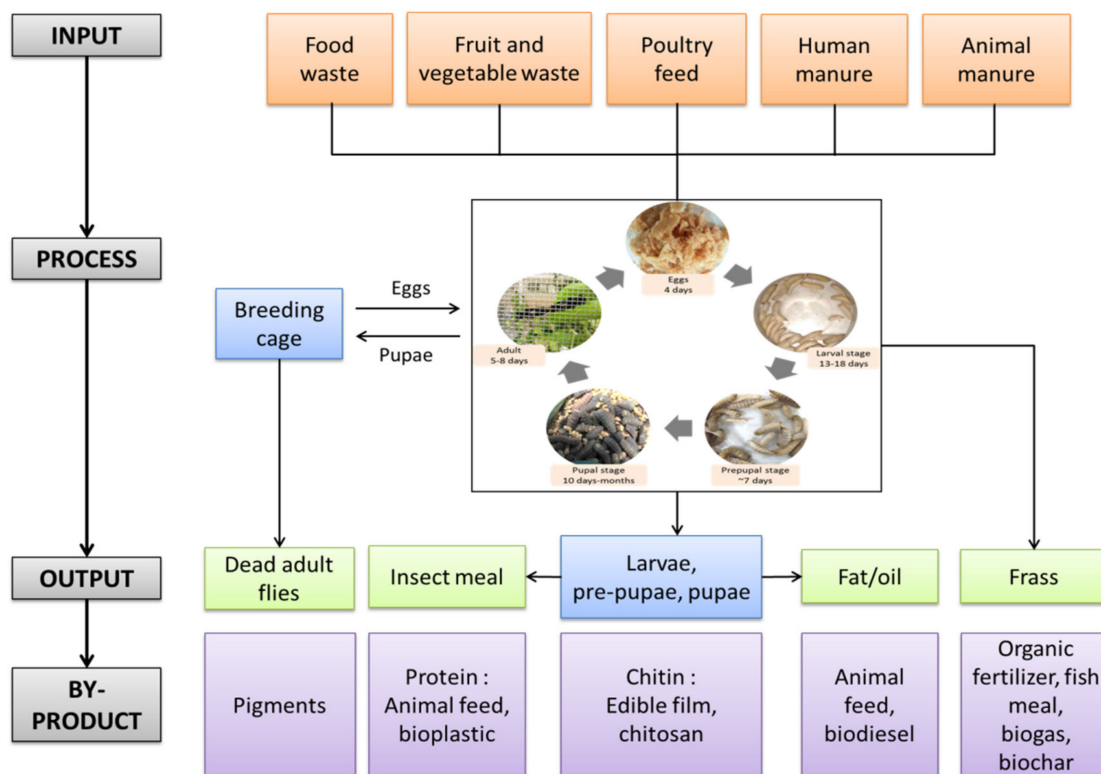


Figure 3. A flow diagram outlining the BSF process.

4.1. Larvae/Prepupae

The larvae are harvested from the larveros before reaching their maximum weight and nutritional value as prepupae. The larvae require some form of post-processing, or product refinement, to ensure they are sanitized before being shipped to the customers. Larvae refinement can include killing, cleaning, sterilizing, and fractioning, for example, to separate the protein and lipid. Dortmans et al. [49] recommended killing the larvae instantly by dipping them in boiling water to sanitize them. Storable products should not have a water and fat content of more than 10%. Sun-drying is a suitable method for reducing the water content, while oil pressing reduces the fat content. BSFL gained attention after the first significant reduction in marine fish stocks that exerted pressure to increase soybean production for use as a fish meal replacement [70]. The decrease in fish meal supply due to exploitation has resulted in a higher price of the fish meal. Hence, there is a need to develop a substitute for the traditional feeding substances using new sustainable sources such as BSFL. BSFL biomass must undergo analysis following the standards set by the Ministry of Agriculture, Livestock and Food Supply (MAPA) in the normative instruction no. 04/2007, concerning best practices to produce quality animal feed in a hygienic condition (MAPA 2007).

BSFL convert various organic wastes into protein and fat-rich biomass [56] that is harmless to animals and is a suitable substitute for the conventional proteins in animal diets [25]. The nutritional composition of the larvae, which is approximately 40% larval protein and 30% larval fat, is dependent on the quality of the substrate fed to the BSFL, particularly the protein and carbohydrate content of the substrate [71]. Table 2 shows that the larvae reared on cow manure (41.2%) have a higher protein composition than those feeding on vegetable and fruit waste (39.9%). The larvae reared on restaurant waste have the highest larval fat content (38.6%). According to Gold et al. [59], animal manure has

higher protein content than fruits and vegetables, and municipal organic solid wastes have the highest lipid content. Several studies have determined the efficiency of BSF meal as a poultry feed [72], which generally shows that BSF meal can replace a significant portion of soybean meal in the poultry diet without affecting the product performance and quality. The studies on several fish species, including channel catfish (*Ictalurus punctatus*) [73], African catfish (*Clarias gariepinus*) [74], Nile tilapia (*Oreochromis niloticus*) [75], rainbow trout (*Oncorhynchus mykiss*) [76], and Atlantic salmon (*Salmo salar*) [77], successfully used BSF meal as a substitute for fish meal. These results indicate that BSF meal could be an alternative protein source for many fish species.

Table 2. Composition of the BSFL fed with different feeding substrates.

Feeding Substrate	BSF Larvae Composition (%)		Reference
	Crude Protein	Fat	
Chicken feed	41.2	33.6	[78]
Restaurant waste	43.1	38.6	[78]
Vegetable and fruit waste	39.9	30.8	[79]
Cow manure	41.2	35.7	[80]
Pig manure	42.8	36.5	[80]

BSFL have high protein content, containing all the essential amino acids required by humans. Despite that, the food safety risk is of the largest concern when introducing BSFL for human consumption, since BSFL are grown on “waste”. Some studies show the feed fed to the BSFL and blanching were discovered to influence the safety of ingesting BSFL, emphasizing the significance of adding adequate decontamination measures, such as blanching, to assure food safety [81]. In general, insect consumption is still a novelty in Western societies because of less knowledge of insect-eating, unlike Thai consumers, who were likely to eat insects because of cultural familiarity [82]. According to surveys of primarily Western consumers, information about the sustainability and nutrition of insects might increase consumers’ confidence to eat a product containing insects, and has the potential to be used as a tool to persuade consumers to try insect-based foods [83]. Thus, many studies investigated BSFL nutrition, which is an early purpose for animal feedstock, and also found that BSFL are relevant for human consumption [81].

In addition to being rich in protein, BSFL also have a high fat content (21–40% of dry mass). BSFL use the fats in their diet during the larval stage as an energy source in their non-feeding adult stage. The fat can be extracted and processed into biodiesel through trans esterification [80]. The idea of using biodiesel (renewable fuel) was popularized globally two or three decades ago. However, several limiting issues, such as the use of edible crops that result in competition with fulfilling food demand and high production cost, necessitate the development of cost-effective biodiesel production alternatives. The conversion of lignocellulosic biomass by BSFL into valuable products is a cost-reduction approach that uses cheaper raw materials to produce biodiesel [84]. BSF fat is high in medium-chain saturated fatty acids and low in polyunsaturated fatty acids, which give the biodiesel a low viscosity and good oxidative stability. Researchers have experimented with various techniques to extract fat from BSF biomass [80,85]. BSF biodiesel has good combustion and fuel efficiency and reduced environmental emissions. Additionally, studies have proven that the quality of biodiesel produced from BSF fat meets international biodiesel standards, including the American Society for Testing and Materials (ASTM D6751) and the EN 14,214 European standard [86].

4.2. Frass

In addition to prepupae/larvae, another main by-product of the BSFL treatment is a valuable residue known as frass, a mixture of BSFL excretion, uneaten substrates, and shed exoskeletons [30,87]. The frass produced by BSFL’s bioconversion of organic waste is approximately 30 to 50% of the initial weight of the feeding substrates [88]. Gao et al. [56]

recommended harvesting the frass when 40–90% of the larvae have transitioned into the prepupae stage. A relatively new idea is to use BSFL frass as an organic fertilizer, since frass has similar characteristics with immature compost [16]. The sustainable use of BSFL frass fertilizer can reduce the over-reliance on costly mineral fertilizers that have harmful effects on the soil and environmental health [89]. The high macronutrient (nitrogen (N), phosphorus (P), potassium (K)), micronutrient, and organic matter content of frass are all readily available for agricultural use [90,91]. However, BSFL frass must first undergo post-processing since the frass is not mature, as BSFL composting occurs very rapidly. Maturation is a critical process to reduce microbial activity in the frass and reduce the competition for oxygen and nitrogen with the soil in which it is applied. Frass maturation can be achieved by vermicomposting and anaerobic digestion [49]. Bulak et al. [92] investigated methane production from insect residues such as BSFL frass. They found that the obtained biomethane potential is similar to the substrate used in biogas plants, demonstrating the feasibility of this method to their utilization and producing a better-quality frass as a fertilizer. However, since biogas plants are quite costly to operate, composting is the most basic approach, where the frass is left on a concrete floor under a roof for three weeks before being applied to the soil.

Post-treatment is necessary to stabilize the BSFL frass. Several studies have shown that using frass as a fertilizer promotes plant growth. There is no significant difference in plant growth when using frass and regular compost [87]. BSFL frass could boost soil organic matter, nitrogen, and phosphorus [93], while the chitin in the frass increases plant resistance against diseases and pathogens. However, it is worth noting that fresh BSFL frass can cause stunted plant growth if applied inappropriately. For example, Alattar et al. [90] attributed the poor corn plant growth to the low porosity of fresh frass, which could have created anaerobic or low-oxygen conditions. Plants need oxygen for respiration and require well-aerated soil for healthy growth. Song et al. [94] studied the effects of further composting on frass quality. They reported that the pak choi fertilized with BSFL frass composted using the forced aeration method produced a higher yield than those fertilized with fresh frass. This study indicates that further composting could increase the quality of BSFL frass used as a fertilizer to encourage food crop development. Therefore, in-depth research on the post-treatment of BSFL frass is needed to improve the quality of frass as an organic fertilizer and make it a better alternative for plant growth than chemical fertilizers.

5. Direction of Further Research

The BSF waste treatment system can be operated on a small scale to focus on a smaller scope. According to the Khazanah Research Institute (KRI), household waste is the primary source of municipal solid waste (MSW) and makes up 44.5% (6.1 million tons) of total solid waste each year. The average amount of uneaten food thrown away by Malaysian households is between 0.5–0.8 kg per day, and this amount is expected to increase with population growth [95]. The increasing amount of food waste could exacerbate the issues related to landfills, such as foul odors, toxic leachate, greenhouse gas emissions, and vermin infestations [96]. One way to resolve these issues is by dealing with the roots of the problems, for example, implementing a BSF system at home to recycle food waste into beneficial products, such as organic fertilizers for home gardens or selling it for profit. The home BSF system does not have transportation costs because it uses uneaten food as a larval feeding substrate. This review has summarized the entire process to help the public understand the operation of the BSF system. However, there are several challenges when operating the BSF system at home, such as odor and limited space. There is a need to design a home-scale BSF system that uses less space, is less time-consuming, and requires little expertise to encourage any interested individual to participate in the mission to reduce food waste. Therefore, further research is needed to determine the optimal conditions for BSF larvae development and an effective waste treatment system that considers the critical environmental factors in residential areas. The user-friendly and well-designed BSF bin, where the larva may be self-harvested and are durable to the Malaysian climate and

environment, should also be noted. The final process of organic waste treatment using BSFL will produce valuable products, generally from larvae or prepupa as an animal feed source, and frass as fertilizer. The larvae can be fed raw to pets and fish, or extracted BSF oil can be used for animal feed formulation. BSF oil is mainly high in saturated fatty acids, especially lauric acid, which is essential in food supplementation for growth performance and antioxidative capacity in livestock [97]. However, the methodology of lauric acid extraction from BSF oil is not fully developed. Hence, it is a need to cover this topic in future studies. In addition to larvae, BSF frass accumulates as a significant byproduct that has the potential to supplement or replace conventional fertilizers. However, BSF bioconversion is a rapid process and produces immature frass. Thus, it is suggested that further research be conducted to investigate the further treatment of frass by using other invertebrates as an composting agent to improve the quality of frass and its fertilizing effects, compared to conventional composting.

6. Conclusions

The researchers believe that the BSF is a “dynamic creature” with many benefits, and that it is an excellent composting agent with minimal environmental impact [98]. BSFL treatment is an evolving technique for converting organic waste into high-value products that are sustainable and commercially viable. Under similar conditions, the nutritional content of organic wastes has the highest impact on process performance [45]. Gold et al. [59] stated that the efficiency of BSFL treatment is dependent on the rate of bioconversion, larval weight, and composition of larval biomass (protein and lipid content), which varies depending on the waste composition and amount of organic waste. The process performance is also dependent on the optimum range of operating parameters stated in this review. The two main by-products from the BSFL waste treatment system are the larvae and prepupae, which can be used as animal feed, while the residue can be used as organic fertilizer. Generally, the BSF organic waste treatment system is a green technology that reduces organic waste, even if the reduction is on a small scale.

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